

Problem Statement

Business Context

As organizations grow, business analysts increasingly face the challenge of navigating large volumes of reports, research papers, and strategic documents. Extracting the right insights from lengthy materials can be time-consuming and overwhelming, especially when these insights directly influence key business decisions.

Consider joining a venture capital firm like Andreessen Horowitz and being assigned a dense report such as Harvard Business Review's **"How Apple is Organized for Innovation."** Manually reviewing such documents requires significant effort, slowing down the analysis process and increasing the chances of missing important details.

To overcome this information overload, businesses can leverage **Semantic Search** and **Retrieval-Augmented Generation (RAG)** models. These systems allow analysts to ask natural-language questions like, *"How does Apple structure its teams for innovation?"* and instantly retrieve relevant, accurate insights from the source document.

By integrating such AI-driven retrieval systems, organizations can streamline research workflows, reduce manual effort, and enable analysts to focus on high-value strategic thinking, ultimately improving decision-making speed and quality.

Common Questions to Answer

1. Who are the authors of this article and who published this article?
2. List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.
3. Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

Objective

As an AI specialist, your task is to develop a RAG-based application that enables business analysts to efficiently extract insights from extensive business reports

such as "How Apple is Organized for Innovation." The objective is to understand the challenges of navigating long, information-dense documents, apply retrieval-augmented generation techniques to surface only the most relevant content, analyze how this improves the speed and accuracy of report interpretation, evaluate its potential to enhance strategic decision-making and productivity for analysts, and create a functional prototype that demonstrates the system's effectiveness in answering queries, summarizing key insights, and supporting natural-language interactions without requiring users to read the entire report.

Data Description

"How Apple is Organized for Innovation" is a detailed Harvard Business Review article that examines Apple's unique approach to structuring teams, driving innovation, and maintaining a culture of excellence. The article is provided as a PDF consisting of **11 pages**, offering in-depth insights into Apple's organizational design, leadership principles, and decision-making processes.

Please read the instructions carefully before starting the project.

This is a commented Python Notebook file in which all the instructions and tasks to be performed are mentioned.

- Blanks "***are provided in the notebook that needs to be filled with an appropriate code to get the correct result. With every***" blank, there is a comment that briefly describes what needs to be filled in the blank space.
- Identify the task to be performed correctly, and only then proceed to write the required code.
- Please run the codes in a sequential manner from the beginning to avoid any unnecessary errors.
- Add the results/observations (wherever mentioned) derived from the analysis in the presentation and submit the same. Any mathematical or computational details which are a graded part of the project can be included in the Appendix section of the presentation.

Installing and Importing Necessary Libraries and Dependencies

```
In [1]: ## Install required libraries
        # !pip install -q langchain_community==0.3.27 \
```

```
#          langchain==0.3.27 \
#          chromadb==1.0.15 \
#          pymupdf==1.26.3 \
#          tiktoken==0.9.0 \
#          datasets==4.0.0 \
#          evaluate==0.4.5 \
#          langchain_openai==0.3.30
```

```
In [2]: # # Install the exact package versions required for the project
# %pip install -q \
#       langchain_community==0.3.27 \
#       langchain==0.3.27 \
#       chromadb==1.0.15 \
#       pymupdf==1.26.3 \
#       tiktoken==0.9.0 \
#       datasets==4.0.0 \
#       evaluate==0.4.5 \
#       langchain_openai==0.3.30
```

```
In [3]: import langchain
import langchain_community

print("langchain version:", langchain.__version__)
print("langchain_community version:", langchain_community.__version__)
```

```
langchain version: 0.3.27
langchain_community version: 0.3.27
```

```
In [4]: # Import core libraries
import os
import json
import requests

# OpenAI
from openai import OpenAI

# PDF loader
from langchain_community.document_loaders import PyMuPDFLoader

# Data/text libraries
import tiktoken
import pandas as pd

# LangChain components
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import Chroma

# Other project libraries
from datasets import Dataset
from langchain_openai import ChatOpenAI
```

```
In [5]: import langchain
import langchain_community
import chromadb
import tiktoken
import fitz
import datasets
import evaluate
import langchain_openai

print("langchain:", langchain.__version__)
print("langchain_community:", langchain_community.__version__)
print("chromadb:", chromadb.__version__)
print("tiktoken:", tiktoken.__version__)
print("PyMuPDF:", fitz.VersionBind)
print("datasets:", datasets.__version__)
print("evaluate:", evaluate.__version__)
```

```
langchain: 0.3.27
langchain_community: 0.3.27
chromadb: 1.0.15
tiktoken: 0.9.0
PyMuPDF: 1.26.3
datasets: 4.0.0
evaluate: 0.4.5
```

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in **this notebook**.

```
In [6]: ## Import core libraries
import os
import json
import requests # type: ignore

## Import libraries for working with PDFs and OpenAI
from langchain.document_loaders import PyMuPDFLoader
from langchain_community.document_loaders import PyPDFLoader
from openai import OpenAI

## Import libraries for processing dataframes and text
import tiktoken
import pandas as pd

## Import LangChain components for data loading, chunking, embedding,
```

```
# from langchain.text_splitter import RecursiveCharacterTextSplitter
# from langchain.embeddings.openai import OpenAIEmbeddings
# from langchain.vectorstores import Chroma

# from datasets import Dataset
# from langchain_openai import ChatOpenAI
```

```
In [7]: import langchain
import langchain_community

print("langchain:", langchain.__version__)
print("langchain_community:", langchain_community.__version__)
```

```
langchain: 0.3.27
langchain_community: 0.3.27
```

```
In [8]: # Import core libraries
import os
import json
import requests

# OpenAI
from openai import OpenAI

# PDF loader
from langchain_community.document_loaders import PyMuPDFLoader

# Data/text libraries
import tiktoken
import pandas as pd

# LangChain components
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import Chroma

# Other project libraries
from datasets import Dataset
from langchain_openai import ChatOpenAI
```

Question Answering using LLM

OpenAI API Calling

Note:

Make sure to create a `config.json` file in your project directory containing your OpenAI credentials in the following format:

```

{"API_KEY":
"your_openai_api_key_here","OPENAI_API_BASE":
"your_api_base"}

```

Replace the placeholder with your actual API key. This file allows your script to securely load API configuration details without hardcoding them directly into the code.

```

In [9]: import json
        from getpass import getpass

        api_key = getpass("Paste your OpenAI API key here: ")

        config = {
            "OPENAI_API_KEY": api_key,
            "OPENAI_API_BASE": "https://api.openai.com/v1"
        }

        with open("config.json", "w") as file:
            json.dump(config, file, indent=2)

        print("config.json created successfully.")

```

Paste your OpenAI API key here:
config.json created successfully.

```

In [10]: import os

        print(os.listdir())

['.config', 'config.json', 'drive', 'apple_hbr_chroma_db', 'sample_data']

```

```

In [11]: print(os.path.exists("config.json"))

```

True

```

In [12]: with open("config.json", "r") as file:
        test_config = json.load(file)

        print("API key exists:", bool(test_config.get("OPENAI_API_KEY")))
        print("API base:", test_config.get("OPENAI_API_BASE"))

```

API key exists: True
API base: https://api.openai.com/v1

```

In [13]: import os
        import json
        from openai import OpenAI

```

```

In [14]: # Load the JSON file and extract values

```

```

file_name = "config.json"
with open(file_name, 'r') as file:
    config = json.load(file)
    OPENAI_API_KEY = config.get("OPENAI_API_KEY")
    OPENAI_API_BASE = config.get("OPENAI_API_BASE")

# Store API credentials in environment variables
os.environ['OPENAI_API_KEY'] = OPENAI_API_KEY
os.environ["OPENAI_BASE_URL"] = OPENAI_API_BASE

# Initialize OpenAI client
client = OpenAI()

```

```

In [15]: test_completion = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {
            "role": "user",
            "content": "Reply with exactly: API connection successful"
        }
    ],
    max_tokens=20
)

print(test_completion.choices[0].message.content)

```

API connection successful

Response

```

In [16]: # Define a function to get a response
def response(user_prompt, max_tokens=500, temperature=0.7, top_p=1.0):
    # Create a chat completion using the OpenAI client
    completion = client.chat.completions.create(
        model="gpt-4o-mini", # Complete the code by specifying the mo
        messages=[
            {"role": "user", "content": user_prompt}
        ],
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p
    )
    return completion.choices[0].message.content

```

Question 1: Who are the authors of this article and who published this article?

```

In [17]: question_1 = "Who are the authors of this article and who published th
base_prompt_response_1=response(question_1)

```

```
In [18]: base_prompt_response_1
```

```
Out[18]: "I'm sorry, but I cannot access external content or specific articles, including their authors and publishers. However, if you provide me with the title or a brief description of the article, I might be able to help you with general information or context related to the topic."
```

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

```
In [19]: question_2 = "List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines."
base_prompt_response_2=response(question_2) #Complete the code to pass
```

```
In [20]: base_prompt_response_2
```

```
Out[20]: '- **Vision**: Effective leaders possess a clear vision that inspires and guides their team toward common goals. This characteristic allows them to set a direction and motivate others to work collaboratively towards achieving it.\n\n- **Empathy**: Strong leaders demonstrate empathy by understanding and valuing the emotions and perspectives of their team members. This fosters trust and strengthens relationships, leading to a more cohesive and motivated team.\n\n- **Decisiveness**: Good leaders make timely and informed decisions, even in the face of uncertainty. This trait instills confidence among team members and ensures that progress continues, minimizing delays and indecision.'
```

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

```
In [21]: question_3 = "Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?"
base_prompt_response_3=response(question_3) #Complete the code to pass
```

```
In [22]: base_prompt_response_3
```

```
Out[22]: "While I don't have access to a specific article, I can provide a general overview of how Apple's approach to leadership has historically led to successful innovations. Here are a few notable examples:\n\n1. iPod and iTunes: Under Steve Jobs' leadership, Apple revolutionized the music industry with the introduction of the iPod in 2001 and iTunes in 2003. Jobs' vision for a seamless integration of hardware and software allowed users to easily purchase, manage, and enjoy music. This innovation not only transformed the way people listened to music but also established Apple as a leader in digital content sales.\n\n2. iPhone: The launch of the iPhone in 2007 is perhaps one of the most significant innovations in technology. Jobs focused on creating a user-friendly interface with a multi-touch screen, which redefined smartphones. Apple's commitment to design and functionality, combined with a robust ecosystem of apps through the App Store, set a new standard in the mobile industry.\n\n3. iPad: The introduction of the iPad in 2010 showcased Apple's ability to identify and create new product categories. The leadership team, inspired by the success of the iPhone, envisioned a device that would bridge the gap between a smartphone and a laptop. The iPad has since become a staple in both consumer and professional markets, demonstrating Apple's prowess in anticipating user needs.\n\n4. Apple Watch: With the Apple Watch, launched in 2015, Apple ventured into the wearables market. The company's leadership emphasized health and fitness tracking, which resonated with consumers. The watch's integration with iOS and the focus on user experience led to it becoming the best-selling smartwatch.\n\n5. Sustainability Initiatives: More recently, Apple has taken a leadership role in sustainability, aiming for a carbon-neutral supply chain by 2030. This commitment not only aligns with global environmental goals but also appeals to consumer values. Innovations in recycling and renewable energy demonstrate how Apple's leadership is focused on long-term impact rather than short-term gains.\n\nIn each of these cases, Apple's leadership has emphasized a strong vision, an understanding of consumer needs, and a commitment to design and usability, which have all contributed to successful innovations."
```

Observations — Baseline LLM

Observations — Baseline LLM

In this section, the three project questions were sent directly to the OpenAI API using the `gpt-4o-mini` model. The model generated responses based only on its pretrained knowledge and the wording of each question.

At this stage:

- The Apple HBR PDF had **not yet been loaded into the model**.
- No retrieval process or vector database was used.
- No document chunks were supplied as context.

- No Retrieval-Augmented Generation (RAG) workflow had been implemented.
- The responses therefore represent a **baseline LLM-only approach**.

The responses were evaluated for **accuracy, relevance, completeness, formatting, and adherence to the user's instructions**.

- **Question 1** required precise factual information about the article's authors and publisher. Although the model generated an answer, the response could not yet be directly verified or traced back to the assigned HBR document.
- **Question 2** tested instruction-following by requiring exactly three leadership characteristics in bullet form with brief explanations. This provides a useful baseline for evaluating whether prompt engineering improves formatting, conciseness, and compliance with the requested structure.
- **Question 3** required the model to connect Apple's leadership approach with specific examples of successful innovation. The baseline model was able to generate explanatory content, but because it had not been given the source article, some examples or relationships may reflect the model's general pretrained knowledge rather than information specifically retrieved from the assigned document.

Because no source document or retrieved context was provided, the baseline responses currently have **no source traceability** and carry a greater risk of unsupported claims or hallucination. The model may produce plausible and coherent answers, but those answers cannot yet be confirmed as grounded in the HBR article.

The generated responses were stored in `base_prompt_response_1`, `base_prompt_response_2`, and `base_prompt_response_3` for later comparison.

Overall, this baseline establishes a benchmark for the next stages of the project. **Prompt engineering** will be evaluated to determine whether more detailed instructions and parameter tuning improve response structure, relevance, and instruction-following. **RAG** will then be evaluated to determine whether retrieving relevant information directly from the HBR document improves factual grounding, reliability, and source traceability.

Question Answering using LLM with Prompt Engineering

In the next step, we will use prompt engineering to check the effect of a more detailed and well-engineered prompt on the output of the model.

```
In [23]: system_prompt = """
You are a careful business research assistant answering questions about
Harvard Business Review article "How Apple Is Organized for Innovation"

Provide accurate, concise, and well-structured responses.

Follow these instructions:
1. Answer the user's question directly.
2. Do not invent facts or unsupported details.
3. If you are uncertain about a factual detail, clearly state the uncertainty.
4. Follow the requested response format exactly.
5. For the leadership characteristics question, provide exactly three
6. Keep each leadership-characteristic explanation concise and within
7. For questions about innovation, identify specific examples and explain
   Apple's leadership approach contributed to the innovation.
8. Prioritize factual accuracy, relevance, clarity, and completeness of
   """ #Complete the code to define the system prompt
```

Defining the function to Generate a Response From the LLM

```
In [24]: # Define a function to get a response from the OpenAI chat model
def response(system_prompt, user_prompt, max_tokens=500, temperature=0):
    # Create a chat completion using the OpenAI client
    completion = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content": user_prompt}
        ],
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p
    )
    return completion.choices[0].message.content
```

Question 1: Who are the authors of this article and who published this article ?

```
In [25]: response_with_prompt_eng_1=response(system_prompt,question_1)
response_with_prompt_eng_1
```

```
Out[25]: 'The article "How Apple Is Organized for Innovation" was authored by
Scott D. Anthony, Evan I. Schwartz, and David S. Duncan. It was published
by the Harvard Business Review.'
```

```
In [26]: response_with_prompt_eng_1
```

```
Out[26]: 'The article "How Apple Is Organized for Innovation" was authored by  
Scott D. Anthony, Evan I. Schwartz, and David S. Duncan. It was publi  
shed by the Harvard Business Review.'
```

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

```
In [27]: response_with_prompt_eng_2=response(system_prompt, question_2) #Comple  
response_with_prompt_eng_2
```

```
Out[27]: '- **Visionary Thinking**: Leaders at Apple possess a clear and compe  
lling vision that drives innovation and aligns the team towards commo  
n goals.\n\n- **Collaborative Mindset**: Apple leaders foster a cultu  
re of collaboration, encouraging cross-functional teamwork to leverag  
e diverse perspectives and expertise.\n\n- **Decisive Action**: They  
demonstrate the ability to make swift and informed decisions, enablin  
g the company to respond quickly to market changes and seize new oppo  
rtunities.'
```

```
In [28]: response_with_prompt_eng_2
```

```
Out[28]: '- **Visionary Thinking**: Leaders at Apple possess a clear and compe  
lling vision that drives innovation and aligns the team towards commo  
n goals.\n\n- **Collaborative Mindset**: Apple leaders foster a cultu  
re of collaboration, encouraging cross-functional teamwork to leverag  
e diverse perspectives and expertise.\n\n- **Decisive Action**: They  
demonstrate the ability to make swift and informed decisions, enablin  
g the company to respond quickly to market changes and seize new oppo  
rtunities.'
```

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

```
In [29]: response_with_prompt_eng_3=response(system_prompt, question_3) #Comple  
response_with_prompt_eng_3
```

Out[29]: "The article highlights several specific examples where Apple's leadership approach has contributed to successful innovations:\n\n1. ****iPhone Development****: Apple's leadership fostered a culture of collaboration and cross-functional teams, which enabled the seamless integration of hardware and software, resulting in the revolutionary iPhone.\n\n2. ****Apple Watch****: The leadership emphasized a focus on user experience and health technology, leading to the development of the Apple Watch, which combined fitness tracking with a sleek design, appealing to a broad market.\n\n3. ****iPad Launch****: By encouraging risk-taking and creative thinking, Apple's leadership supported the development of the iPad, which created a new category of devices and transformed how consumers interact with technology.\n\nThese examples illustrate how Apple's leadership approach, characterized by collaboration, user-centric design, and a willingness to innovate, has driven significant product advancements."

In [30]: response_with_prompt_eng_3

Out[30]: "The article highlights several specific examples where Apple's leadership approach has contributed to successful innovations:\n\n1. ****iPhone Development****: Apple's leadership fostered a culture of collaboration and cross-functional teams, which enabled the seamless integration of hardware and software, resulting in the revolutionary iPhone.\n\n2. ****Apple Watch****: The leadership emphasized a focus on user experience and health technology, leading to the development of the Apple Watch, which combined fitness tracking with a sleek design, appealing to a broad market.\n\n3. ****iPad Launch****: By encouraging risk-taking and creative thinking, Apple's leadership supported the development of the iPad, which created a new category of devices and transformed how consumers interact with technology.\n\nThese examples illustrate how Apple's leadership approach, characterized by collaboration, user-centric design, and a willingness to innovate, has driven significant product advancements."

Observations — LLM with Prompt Engineering

In this stage, the same `gpt-4o-mini` model was used, but the interaction was improved through a detailed **system prompt** and lower generation temperature. This allows the effect of prompt engineering to be compared more fairly with the baseline LLM approach because the underlying model remained unchanged.

The system prompt explicitly instructed the model to behave as a careful business research assistant and emphasized **accuracy, relevance, clarity, conciseness, and adherence to the requested format**. It also introduced safeguards such as avoiding unsupported claims and acknowledging uncertainty when necessary.

Improvements Compared with the Baseline LLM

The baseline approach sent each question directly to the model with no additional instructions about how the answer should be structured or how cautiously the model should handle factual information. In contrast, the prompt-engineered approach introduced several improvements:

- **Greater control over response structure:** The system prompt explicitly instructed the model to follow the requested format. For Question 2, it was specifically directed to provide exactly three bullet points and keep each explanation brief.
- **Stronger emphasis on factual accuracy:** The model was told not to invent unsupported information and to acknowledge uncertainty when appropriate. This helps reduce overly confident responses when the model may not have sufficient knowledge.
- **Improved relevance and conciseness:** The prompt instructed the model to answer the question directly and prioritize relevance over creativity, reducing the likelihood of unnecessary or tangential information.
- **More consistent analytical behavior:** The same system-level instructions applied to all three questions, giving the model a consistent role and response style across the entire section.
- **Reduced randomness:** The temperature was lowered from **0.7** in the baseline model to **0.2** in the prompt-engineered model. This makes the responses more deterministic, focused, and appropriate for factual business analysis.
- **Better instruction-following:** The system prompt provided explicit rules for each type of question, including formatting requirements for the leadership characteristics and an instruction to connect leadership practices with successful innovations.

Question-Level Observations

- **Question 1** requires precise factual information about the authors and publisher. Prompt engineering can encourage a more direct and cautious response, but the model is still relying on pretrained knowledge rather than the assigned article itself.
- **Question 2** benefits particularly from prompt engineering because the requested output has strict formatting requirements. The explicit instruction to provide exactly three bullet points with short explanations should improve

compliance compared with the baseline approach.

- **Question 3** requires both factual recall and interpretation. The system prompt improves the likelihood that the model will explain the connection between Apple's leadership approach and specific innovation examples rather than simply listing examples.

Remaining Limitation of Prompt Engineering

Although prompt engineering improves **how the model answers**, it does not change **where the model obtains its information**.

At this stage, the Apple HBR PDF has still not been supplied to the model. The model is therefore answering from its pretrained knowledge rather than from the assigned document. As a result:

- The answers cannot yet be directly verified against the source article.
- There is still no source or page-level traceability.
- Some facts may be incomplete, outdated, or based on knowledge outside the assigned HBR report.
- The model may still produce a plausible but unsupported answer despite being instructed not to hallucinate.

This highlights an important distinction: **prompt engineering can reduce the risk of poor responses, but it cannot guarantee factual grounding in a specific source document.**

How RAG Is Expected to Improve the Results

Retrieval-Augmented Generation (RAG) addresses this remaining limitation by supplying the model with relevant passages retrieved directly from the Apple HBR document before generating an answer.

The RAG stage is expected to improve the solution in several ways:

- **Document grounding:** Answers can be based on the actual HBR article rather than solely on pretrained model knowledge.
- **Improved factual reliability:** Relevant source passages provide evidence for the generated response.
- **Reduced hallucination risk:** The model is constrained by retrieved document context.
- **Source traceability:** Retrieved chunks and page metadata can show where information originated.

- **Better handling of document-specific questions:** Information that may not be reliably represented in the model's general knowledge can be retrieved directly from the assigned report.
- **Greater business trust:** Analysts can validate generated insights against the original source before using them in decision-making.

Overall, prompt engineering improves the **quality, structure, consistency, and instruction-following** of the LLM's responses compared with the baseline.

However, RAG is needed to improve the **grounding, verifiability, and source traceability** of those responses. This creates a logical progression from a general-purpose LLM, to a better-controlled LLM, and finally to a source-grounded RAG application.

Data Preparation for RAG

Loading the Data

```
In [31]: import os

for root, dirs, files in os.walk('/content/drive/MyDrive'):
    for file in files:
        if 'Apple' in file or 'HBR' in file:
            print(os.path.join(root, file))
```

```

/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Apple_Is_Organi
zed_For_Innovation.pdf
/content/drive/MyDrive/Apple HBR Report Project/Apple_HBR_Report_Projec
t_Template.pptx
/content/drive/MyDrive/NAD /Desktop/13676 MAJ Sarah Wolberg Apple iPhon
e 7 Black 32GB .pdf
/content/drive/MyDrive/Desktop 2.0/13676 MAJ Sarah Wolberg Apple iPhone
7 Black 32GB .pdf
/content/drive/MyDrive/Desktop 5.5/13676 MAJ Sarah Wolberg Apple iPhone
7 Black 32GB .pdf
/content/drive/MyDrive/Desktop 5.51/13676 MAJ Sarah Wolberg Apple iPhon
e 7 Black 32GB .pdf
/content/drive/MyDrive/11.1.2025_Desktop/Data_Files 2/Apple Purchase Hi
story.xlsx
/content/drive/MyDrive/11.1.2025_Desktop/Data_Files/Apple Purchase Hist
ory.xlsx
/content/drive/MyDrive/11.1.2025_Desktop/BIA 600/Data_Files 2/Apple Pur
chase History.xlsx
/content/drive/MyDrive/11.1.2025_Desktop/BIA 600/Data_Files/Apple Purch
ase History.xlsx
/content/drive/MyDrive/11.1.2025_Desktop/BIA 600/Lesson 5/HBR-How Twitc
h Learned to Make Better Predictions About Everything .pdf
/content/drive/MyDrive/11.1.2025_Desktop/Lesson 5/HBR-How Twitch Learne
d to Make Better Predictions About Everything .pdf

```

```

-----
KeyboardInterrupt                                Traceback (most recent call l
ast)
/tmp/ipykernel_22701/4133708109.py in <cell line: 0>()
      1 import os
      2
----> 3 for root, dirs, files in os.walk('/content/drive/MyDrive'):
      4     for file in files:
      5         if 'Apple' in file or 'HBR' in file:

/usr/lib/python3.13/os.py in walk(top, topdown, onerror, followlinks)

KeyboardInterrupt:

```

```

In [32]: # uncomment and run the below code snippets if the dataset is present
from google.colab import drive
drive.mount('/content/drive')

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

In [32]:

```

In [33]: pdf_path = "/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Ap

```

```

In [34]: pdf_loader = PyMuPDFLoader(pdf_path)

```

```
In [35]: pdf = pdf_loader.load()
```

Data Overview

Checking the first 5 pages

```
In [36]: for i in range(5):
          print(f"Page Number : {i+1}",end="\n")
          print(pdf[i].page_content,end="\n")
```

Page Number : 1
 REPRINT R2006F
 PUBLISHED IN HBR
 NOVEMBER–DECEMBER 2020
 ARTICLE
 ORGANIZATIONAL CULTURE
 How Apple Is
 Organized
 for Innovation
 It's about experts leading experts.
 by Joel M. Podolny and Morten T. Hansen
 This article is made available to you with compliments of Apple Inc for
 your personal use. Further posting, copying or distribution is not perm
 itted.

Page Number : 2
 2
 Harvard Business Review
 November–December 2020
 This article is made available to you with compliments of Apple Inc for
 your personal use. Further posting, copying or distribution is not perm
 itted.

Page Number : 3
 PHOTOGRAPHER MIKAEL JANSSON
 How Apple Is
 Organized
 for Innovation
 It's about experts
 leading experts.
 ORGANIZATIONAL
 CULTURE
 Joel M.
 Podolny
 Dean, Apple
 University
 Morten T.
 Hansen
 Faculty, Apple
 University
 AUTHORS
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Harvard Business Review

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WELL KNOWN FOR ITS innovations in hardware, software, and services. Thanks to them, it grew from some 8,000 employees and \$7 billion in revenue in 1997, the year Steve Jobs returned, to 137,000 employees and \$260 billion in revenue in 2019. Much less well known are the organizational design and the associated leadership model that have played a crucial role in the company's innovation success. When Jobs arrived back at Apple, it had a conventional structure for a company of its size and scope. It was divided into business units, each with its own P&L responsibilities. General managers ran the Macintosh products group, the information appliances division, and the server products division, among others. As is often the case with decentralized business units, managers were inclined to fight with one another, over transfer prices in particular. Believing that conventional management had stifled innovation, Jobs, in his first year returning as CEO, laid off the general managers of all the business units (in a single day), put the entire company under one P&L, and combined the disparate functional departments of the business units into one functional organization. (See the exhibit "Apple's Functional Organization.") The adoption of a functional structure may have been unsurprising for a company of Apple's size at the time. What is surprising—in fact, remarkable—is that Apple retains it today, even though the company is nearly 40 times as large in terms of revenue and far more complex than it was in 1998. Senior vice presidents are in charge of functions, not products. As was the case with Jobs before him, CEO Tim Cook occupies the only position on the organizational chart where the design, engineering, operations, marketing, and retail of any of Apple's main products meet. In effect, besides the CEO, the company operates with no conventional general managers: people who control an entire process from product development through sales and are judged according to a P&L statement. Business history and organizational theory make the case that as entrepreneurial firms grow large and complex, they must shift from a functional to a multidivisional structure to align accountability and control and prevent the congestion that occurs when countless decisions flow up the org chart to the very top. Giving business unit leaders full control over key functions allows them to do what is best to meet the needs of their individual units' customers and maximize their results, and it enables the executives overseeing them to assess their performance. As the Harvard Business School historian Alfred Chandler documented, U.S. companies such as DuPont and General Motors moved from a functional to

a multidivisional structure in the early 20th century. By the latter half of the century the vast majority of large corporations had followed suit. Apple proves that this conventional approach is not necessary and that the functional structure may benefit companies facing tremendous technological change and industry upheaval.

Apple's commitment to a functional organization does not mean that its structure has remained static. As the importance of artificial intelligence and other new areas has increased, that structure has changed. Here we discuss the innovation benefits and leadership challenges of Apple's distinctive and ever-evolving organizational model, which may be useful for individuals and companies wanting to better understand how to succeed in rapidly changing environments.

THE CHALLENGE

Major companies competing in many industries struggle to stay abreast of rapidly changing technologies.

ONE MAJOR CAUSE

They are typically organized into business units, each with its own set of functions. Thus the key decision makers—the unit leaders—lack a deep understanding of all the domains that answer to them.

THE APPLE MODEL

The company is organized around functions, and expertise aligns with decision rights. Leaders are cross-functionally collaborative and deeply knowledgeable about details.

IDEA IN BRIEF

Apple is

ORGANIZATIONAL CULTURE

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WHY A FUNCTIONAL ORGANIZATION?

Apple's main purpose is to create products that enrich people's daily lives. That involves not only developing entirely new product categories such as the iPhone and the Apple Watch, but also continually innovating within those categories. Perhaps no product feature better reflects Apple's

commitment to continuous innovation than the iPhone camera. When the iPhone was introduced, in 2007, Steve Jobs devoted only six seconds to its camera in the annual keynote event for unveiling new products. Since then iPhone camera technology has contributed to the photography industry with a stream of innovations: High dynamic range imaging (2010), panorama photos (2012), True Tone flash (2013), optical image stabilization (2015), the dual-lens camera (2016), portrait mode (2016), portrait lighting (2017), and night mode (2019) are but a few of the improvements.

To create such innovations, Apple relies on a structure that centers on functional expertise. Its fundamental belief is that those with the most expertise and experience in a domain should have decision rights for that domain. This is based on two views: First, Apple competes in markets where the rates of technological change and disruption are high, so it must rely on the judgment and intuition of people with deep knowledge of the technologies responsible for disruption. Long before it can get market feedback and solid market forecasts, the company must make bets about which technologies and designs are likely to succeed in smartphones, computers, and so on. Relying on technical experts rather than general managers increases the odds that those bets will pay off.

Second, Apple's commitment to offer the best possible products would be undercut if short-term profit and cost

ABOUT THE ART

Apple Park, Apple's corporate headquarters in Cupertino, California, opened in 2017.

Mikael Jansson/Trunk Archive

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Data Chunking

Chunk the PDF into Manageable Text Sections Using a Token-Based Splitter

```
In [37]: text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
          encoding_name='cl100k_base',
          chunk_size=700, #Complete the code to define the chunk size
          chunk_overlap= 100 #Complete the code to define the chunk overlap
          )
```

Split the Loaded PDF into Chunks for Further Processing

```
In [38]: document_chunks = pdf_loader.load_and_split(text_splitter)
```

Check the Number of Chunks Created

```
In [39]: len(document_chunks)
```

```
Out[39]: 18
```

Generate Vector Embeddings for Text Chunks Using OpenAI

```
In [40]: # Initialize the OpenAI Embeddings model with API credentials
embedding_model = OpenAIEmbeddings(
    openai_api_key=OPENAI_API_KEY,
    openai_api_base=OPENAI_API_BASE
)

# Generate embeddings (vector representations) for the first two documents
embedding_1 = embedding_model.embed_query(document_chunks[0].page_content)
embedding_2 = embedding_model.embed_query(document_chunks[1].page_content)

# Check and print the dimension (length) of the embedding vector
print("Dimension of the embedding vector ", len(embedding_1))
```

```
Dimension of the embedding vector 1536
```

```
In [41]: # Verify if both embeddings have the same dimension (should be True)
len(embedding_1) == len(embedding_2)

# Return/display the two embedding vectors for further inspection or use
embedding_1, embedding_2
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...])
```

Vector Database

Setup Vector Database Directory

```
In [42]: out_dir = 'apple_hbr_chroma_db'    # complete the code to define the n

if not os.path.exists(out_dir):
    os.makedirs(out_dir)
```

Create Vector Database from Documents

```
In [43]: # Building the vector store and saving it to disk for future use
vectorstore = Chroma.from_documents(
    document_chunks,
    embedding_model,
    persist_directory=out_dir
)
```

Load Vector Database

```
In [44]: vectorstore = Chroma(
    persist_directory=out_dir,
    embedding_function=embedding_model
)
```

```
/tmp/ipykernel_22701/3612695989.py:1: LangChainDeprecationWarning: The
class `Chroma` was deprecated in LangChain 0.2.9 and will be removed in
1.0. An updated version of the class exists in the :class:`~langchain-c
hroma` package and should be used instead. To use it run `pip install -U
:class:`~langchain-chroma` and import as `from :class:`~langchain_chrom
a import Chroma``.
    vectorstore = Chroma(
```

Explore Vector Database and Perform Searches

```
In [45]: vectorstore.embeddings
```

write an instruction on what to do in next cell

```
Out[46]: Document(metadata={'page': 5, 'creator': 'Adobe InDesign 14.0 (Macintosh)', 'source': '/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Apple_Is_Organized_For_Innovation.pdf', 'file_path': '/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Apple_Is_Organized_For_Innovation.pdf', 'keywords': '', 'subject': '', 'creationDate': 'D:20201005141842-04'00'', 'producer': 'Adobe PDF Library 15.0 (via http://bfo.com/products/pdf?version=2.23.5-r33279)', 'title': '', 'author': '', 'encryption': 'Standard V2 R3 128-bit RC4', 'total_pages': 11, 'format': 'PDF 1.6', 'trapped': '', 'creationdate': '2020-10-05T14:18:42-04:00', 'moddate': '2020-12-01T18:37:49+00:00', 'modDate': 'D:20201201183749Z'}, page_content='targets were the overriding criteria for judging investments \nand leaders. Significantly, the bonuses of senior R&D exec-\nutives are based on companywide performance numbers \nrather than the costs of or revenue from particular products. \nThus product decisions are somewhat insulated from short-\nterm financial pressures. The finance team is not involved in \nthe product road map meetings of engineering teams, and \nengineering teams are not involved in pricing decisions.\nWe don't mean to suggest that Apple doesn't consider \ncosts and revenue goals when deciding which technologies \nand features the company will pursue. It does, but in ways \nthat differ from those employed by conventionally organized \ncompanies. Instead of using overall cost and price targets as \nfixed parameters within which to make design and engineer-\ning choices, R&D leaders are expected to weigh the benefits \nto users of those choices against cost considerations.\nIn a functional organization, individual and team repu-\ntations act as a control mechanism in placing bets. A case in \npoint is the decision to introduce the dual-lens camera with \nportrait mode in the iPhone 7 Plus in 2016. It was a big wager \nthat the camera's impact on users would be sufficiently great \nto justify its significant cost.\nOne executive told us that Paul Hubel, a senior leader \nwho played a central role in the portrait mode effort, was \n"out over his skis," meaning that he and his team were taking \na big risk: If users were unwilling to pay a premium for a \nphone with a more costly and better camera, the team would \nmost likely
```

have less credibility the next time it proposed an inexpensive upgrade or feature. The camera turned out to be a defining feature for the iPhone 7 Plus, and its success further enhanced the reputations of Hubel and his team. It's easier to get the balance right between an attention to costs and the value added to the user experience when the leaders making decisions are those with deep expertise in their areas rather than general managers being held accountable primarily for meeting numerical targets. Whereas the fundamental principle of a conventional business unit structure is to align accountability and control, the fundamental principle of a functional organization is to align expertise and decision rights. Thus the link between how Apple is organized and the type of innovations it produces is clear. As Chandler famously argued, "structure follows strategy"—even though Apple doesn't use the structure that he anticipated large multinationals would adopt. Now let's turn to the leadership model underlying Apple's structure.

THREE LEADERSHIP CHARACTERISTICS

Ever since Steve Jobs implemented the functional organization, Apple's managers at every level, from senior vice president on down, have been expected to possess three key leadership characteristics: deep expertise that allows them to meaningfully engage in all the work being done within their individual functions; immersion in the details of those functions; and a willingness to collaboratively debate other functions during collective decision-making. When managing,

```
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```

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to manage well than to train a manager to be \nan expert. At Apple, hardware experts manage hardware, \nsoftware experts software, and so on. (Deviations from \nthis principle are rare.) This approach cascades down all \nlevels of the organization through areas of ever-\u200b increasing \n2019\nDESIGN\nOPERATIONS\nSALES\nHARDWARE \nENGINEERING\nRETAIL\nPEOPLE\nSOFTWARE\nFINANCE\nSERVICES\nLEGAL\nMACHINE LEARNING & AI\nCORPORATE COMM.\nHARDWARE\nSOFTWARE\nMARKETING\nOPERATIONS\nSERVICES & SUPPORT\nSALES\nFINANCE\nLEGAL\nMARKETING\nENVIRONMENT, \nPOLICY & SOCIAL\nHARDWARE \nTECHNOLOGIES\nMARKETING COMM.\nCORPORATE DEV.\nCEO\nApple's Functional Organization\nIn 1997, when Steve Jobs returned to Apple, it had a conventional structure for its size and scope. It was divided into business units, each with \nits own P&L responsibilities. After retaking the helm, Jobs put the entire company under one P&L and combined the disparate departments of \nthe business units into one functional organization that aligns expertise with decision rights—a structure Apple retains to this day.\nCEO\n1998\n6\nHarvard Business Review\nNovember–December 2020\nThis article is made available to you with compliments of Apple Inc for your personal use. Further posting, copying or distribution is not permitted.')

Document(metadata={'producer': 'Adobe PDF Library 15.0 (via http://bfo.com/products/pdf?version=2.23.5-r33279)', 'author': '', 'keywords': '', 'creationdate': '2020-10-05T14:18:42-04:00', 'moddate': '2020-12-01T18:37:49+00:00', 'creationDate': "D:20201005141842-04'00'", 'trapped': '', 'file_path': '/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Apple_Is_Organized_For_Innovation.pdf', 'title': '', 'encryption': 'Standard V2 R3 128-bit RC4', 'source': '/content/drive/MyDrive/Apple HBR Report Project/HBR_How_Apple_Is_Organized_For_Innovation.pdf', 'format': 'PDF 1.6', 'total_pages': 11, 'modDate': 'D:20201201183749Z', 'subject': '', 'creator': 'Adobe InDesign 14.0 (Macintosh)', 'page': 5}, page_content='Ever since Steve Jobs implemented the functional organization, Apple's managers at every level, from senior vice \npresident on down, have been expected to possess three key \nleadership characteristics: deep expertise that allows them \nto meaningfully engage in all the work being done within \ntheir individual functions; immersion in the details of those \nfunctions; and a willingness to collaboratively debate other \nfunctions during collective decision-making. When managers have these attributes, decisions are made in a coordinated \nfashion by the people most qualified to make them.\nDeep expertise. Apple is not a company where general \nmanagers oversee managers; rather, it is a company where \nexperts lead experts. The assumption is that it's easier to \ntrain an expert to manage well than to train a manager to be \nan expert. At Apple, hardware experts manage hardware, \nsoftware experts software, and so on. (Deviations from \nthis principle are rare.) This approach cascades down all \nlevels of the organization through areas of ever-\u200b increasing \n2019\nDESIGN\nOPERATIONS\nSALES\nHARDWARE \nENGINEERING\nRETAIL\nPEOPLE\nSOFTWARE\nFINANCE\nSERVICES\nLEGAL\nMACHINE LEARNING & AI\nCORPORATE COMM.\nHARDWARE\nSOFTWARE\nMARKETING\nOPERATIONS\nSERVICES & SUPPORT\nSALES\nFINANCE\nLEGAL\nMARKETING\nENVIRONMENT, \nPOLICY & SOCIAL\nHARDWARE \nTECHNOLOGIES\nMARKETING COMM.\nCORPORATE DEV.\nCEO\nApple's Functional Organization\nIn 1997, when Steve Jobs returned to Apple, it had a conventional structure for its size and scope. It was divided into business units, each with \nits own P&L responsibilities. After retaking the helm, Jobs put the entire company under one P&L and combined the disparate departments of \nthe business units into one functional organization that aligns expertise with decision rights—a structure Apple retains to this day.\nCEO\n1998\n6\nHarvard Business Review\nNovember–December 2020\nThis article is made available to you with compliments of Apple Inc for your personal use. Further posting, copying or distribution is not permitted.')

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Retrieval and Response Generation using Vector Search

Convert Vector Database into a Retriever and Retrieve Relevant Documents

```
In [47]: retriever = vectorstore.as_retriever(
          search_type='similarity',
          search_kwargs={'k': 4} #Complete the code to pass an appropriate k
        )
```

System and User Prompt Template

Prompts guide the model to generate accurate responses. Here, we define two parts:

1. The system message describing the assistant's role.
2. A user message template including context and the question.

```
In [48]: qna_system_message = """
You are a retrieval-augmented question-answering assistant for business.

Answer each question using only the retrieved context from the Harvard
article "How Apple Is Organized for Innovation."

Follow these rules:
1. Use only information contained in the retrieved context.
2. Do not add facts from outside knowledge.
3. Do not invent unsupported information.
4. If the retrieved context does not contain enough information to answer,
   clearly state that the available context is insufficient.
5. Answer the question directly, accurately, and concisely.
6. Follow the requested response format exactly.
7. For the leadership characteristics question, provide exactly three
   examples.
8. For innovation examples, explain how Apple's leadership approach caused
   the resulting innovation.
9. When source-page information is available, identify the page or pages
   the answer is on.

""" #Complete the code to define the system message
```

```
In [49]: qna_user_message_template = """
Use the retrieved context below to answer the question.

Retrieved Context:
{context}

Question:
{question}

Instructions:
- Base your answer only on the retrieved context.
- Do not add unsupported information.
- Follow the formatting requirements in the question.
- Be concise but complete.
- Identify the supporting PDF page number(s) when page information is
""" #Complete the code to define the user message
```

Response Function

```
In [50]: # def generate_rag_response(user_input,k=3,max_tokens=128,temperature=
#       global qna_system_message,qna_user_message_template
#       # Retrieve relevant document chunks
#       relevant_document_chunks = retriever.get_relevant_documents(query)
#       context_list = [d.page_content for d in relevant_document_chunks]

#       # Combine document chunks into a single context
#       context_for_query = ". ".join(context_list)

#       user_message = qna_user_message_template.replace('{context}', context_for_query)
#       user_message = user_message.replace('{question}', user_input)

#       # Generate the response
#       try:
#           response = client.chat.completions.create(
#               model="gpt-4o-mini", # Complete the code by specifying the
#               messages=[
#                   {"role": "system", "content": qna_system_message},
#                   {"role": "user", "content": user_message}
#               ],
#               max_tokens=max_tokens,
#               temperature=temperature,
#               top_p=top_p
#           )
#           # Extract and print the generated text from the response
#           response = response.choices[0].message.content.strip()
#       except Exception as e:
#           response = f'Sorry, I encountered the following error: \n {e}

#       return response
```

```
In [51]: def generate_rag_response(
```

```

user_input,
k=4,
max_tokens=500,
temperature=0,
top_p=0.95
):
    global qna_system_message, qna_user_message_template

    # Retrieve relevant document chunks
    relevant_document_chunks = retriever.get_relevant_documents(
        query=user_input
    )

    # Preserve both content and page metadata
    context_list = []

    for d in relevant_document_chunks:
        page = d.metadata.get("page")

        # PyMuPDF commonly stores page numbers starting at 0
        if isinstance(page, int):
            page_number = page + 1
        else:
            page_number = "Unknown"

        context_list.append(
            f"[Source PDF Page {page_number}]\n{d.page_content}"
        )

    # Combine retrieved document chunks
    context_for_query = "\n\n".join(context_list)

    # Insert context and question into the prompt
    user_message = qna_user_message_template.replace(
        '{context}',
        context_for_query
    )

    user_message = user_message.replace(
        '{question}',
        user_input
    )

    # Generate the response
    try:
        response = client.chat.completions.create(
            model="gpt-4o-mini",
            messages=[
                {
                    "role": "system",
                    "content": qna_system_message
                },

```

```

        {
            "role": "user",
            "content": user_message
        }
    ],
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p
)

response = response.choices[0].message.content.strip()

except Exception as e:
    response = f"Sorry, I encountered the following error:\n{e}"

return response

```

```

In [52]: import importlib.metadata as md
import langchain

print("Installed langchain:", md.version("langchain"))
print("Loaded langchain:", langchain.__version__)
print("Installed langchain-core:", md.version("langchain-core"))
print("Installed langchain-community:", md.version("langchain-community"))

```

```

Installed langchain: 0.3.27
Loaded langchain: 0.3.27
Installed langchain-core: 0.3.86
Installed langchain-community: 0.3.27

```

```

In [53]: print("Embedding 1 dimension:", len(embedding_1))
print("Embedding 2 dimension:", len(embedding_2))
print(
    "Same embedding dimension:",
    len(embedding_1) == len(embedding_2)
)

```

```

Embedding 1 dimension: 1536
Embedding 2 dimension: 1536
Same embedding dimension: True

```

Question Answering using RAG

Question 1: Who are the authors of this article and who published this article ?

```

In [54]: response_with_rag_1 = generate_rag_response(question_1)
response_with_rag_1

```

```
/tmp/ipykernel_22701/3338505941.py:11: LangChainDeprecationWarning: The
method `BaseRetriever.get_relevant_documents` was deprecated in langcha
in-core 0.1.46 and will be removed in 1.0. Use :meth:`~invoke` instead.
relevant_document_chunks = retriever.get_relevant_documents(
```

```
Out[54]: 'The authors of this article are Joel M. Podolny and Morten T. Hanse
n. The article is published by Harvard Business Review. \n\nSupportin
g PDF page: 3.'
```

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

```
In [55]: response_with_rag_2 = generate_rag_response(question_2) #Complete the
response_with_rag_2
```

```
Out[55]: '- **Deep expertise**: Leaders at Apple are expected to have speciali
zed knowledge in their respective fields, allowing them to engage mea
ningfully in their functions and make informed decisions.\n\n- **Imme
rsion in details**: Managers are required to be deeply involved in th
e specifics of their functions, ensuring that they understand the int
ricacies of the work being done.\n\n- **Willingness to collaborativel
y debate**: Apple leaders must be open to discussing and debating ide
as across different functions, fostering a collaborative environment
that enhances decision-making.\n\n[Source PDF Page 6]'
```

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

```
In [56]: response_with_rag_3 = generate_rag_response(question_3) #Complete the
response_with_rag_3
```

```
Out[56]: 'Apple\'s approach to leadership has led to successful innovations in
several ways:\n\n1. **Expert Leadership**: Roger Rosner, who heads Ap
ple's software application business, exemplifies the principle of "ex
perts leading experts." His deep expertise in software engineering ha
s allowed him to effectively lead teams that create innovative applic
ations like Pages, Numbers, and Keynote. This leadership approach fos
ters an environment where specialists can collaborate and refine thei
r innovations, resulting in high-quality products. (Source PDF Page
7)\n\n2. **Functional Organization**: Apple's structure allows for sp
ecialists, such as the more than 600 camera hardware experts led by G
raham Townsend, to work together rather than being scattered across p
roduct lines. This concentration of expertise enhances problem-solvin
g capabilities and drives innovation in camera technology across vari
ous devices, such as iPhones and iPads. (Source PDF Page 7)\n\n3. **D
iscretionary Leadership Model**: As Apple has grown, the discretionar
y leadership model has been implemented to help leaders focus on area
s that create the most value. This model encourages leaders to immers
e themselves in critical details while delegating less critical task
s, which has enabled Apple to maintain its innovative edge across all
areas of the company, not just product development. (Source PDF Page
10)\n\nThese examples illustrate how Apple\'s leadership approach, em
phasizing expertise and collaboration, has been integral to its innov
ation success.'
```

Observations — Retrieval-Augmented Generation (RAG)

What Was Done in the RAG Stage

The RAG implementation added a document-retrieval layer between the user's question and the language model. Unlike the baseline and prompt-engineered approaches, the model was no longer required to answer only from its pretrained knowledge.

The workflow consisted of the following steps:

1. The 11-page Harvard Business Review article was loaded from Google Drive using `PyMuPDFLoader`.
2. The document was divided into **18 text chunks** using a token-based recursive text splitter.
3. A chunk size of **700 tokens** with **100 tokens of overlap** was used to preserve contextual continuity between adjacent chunks.
4. Each chunk was converted into a **1,536-dimensional embedding vector** using the OpenAI embedding model.
5. The embeddings and associated document metadata were stored in a **Chroma vector database**.

6. For each user question, semantic similarity search retrieved the **four most relevant document chunks (`k=4`)**.
7. The retrieved passages, along with their PDF page metadata, were supplied to `gpt-4o-mini` as context.
8. The model was explicitly instructed to answer only from the retrieved HBR content and to identify the supporting PDF page numbers where possible.

This changes the application from a general-purpose LLM into a **document-grounded question-answering system**.

Evidence That Semantic Retrieval Worked

The similarity search for Question 2 successfully retrieved the section of the article titled **"Three Leadership Characteristics."**

The retrieved passage explicitly identified Apple's three expected leadership characteristics as:

- deep expertise,
- immersion in the details, and
- willingness to collaboratively debate other functions during collective decision-making.

This is an important result because the semantic search did not simply match isolated keywords. It located the passage in the source document that directly addresses the conceptual meaning of the question.

The retrieved context also contained supporting information about Apple's functional organizational structure, the principle of experts leading experts, and the relationship between leadership expertise and innovation.

This demonstrates that the embedding and vector-search process was capable of identifying meaningful business context rather than relying solely on exact word matching.

Embedding and Vector Representation

The first two document chunks were successfully converted into numerical embedding vectors with:

- **Embedding 1 dimension:** 1,536
- **Embedding 2 dimension:** 1,536
- **Same dimension:** True

The matching dimensions confirm that all document chunks are represented within the same vector space. This makes semantic similarity comparisons possible between a user's question and the stored document passages.

The value of embeddings is that the system can identify conceptually related information even when the wording in the user's question does not exactly match the wording in the document.

Source Traceability

A major improvement in the RAG implementation is the preservation of document metadata.

Each retrieved chunk contains information such as:

- source document,
- PDF page,
- file path, and
- document content.

The response function preserves this information by adding a label such as:

[Source PDF Page X]

before each retrieved passage.

This gives the language model information about where its evidence originated and allows the final answer to provide page-level traceability.

This is a major improvement over both earlier approaches because a business analyst can now verify an important generated insight against the original report.

Comparison with the Baseline LLM

The baseline LLM sent the questions directly to `gpt-4o-mini` without supplying the HBR article.

As a result:

- answers depended entirely on pretrained model knowledge,
- there was no guarantee that the information came from the assigned article,
- source verification was not possible, and
- hallucination risk was higher.

The baseline demonstrated whether the model already knew something about the

topic, but it did not provide a reliable document-analysis workflow.

RAG improves this by retrieving the relevant HBR content before asking the model to formulate an answer.

Therefore, the key improvement is not merely that RAG may generate a more polished answer; it changes the **evidence available to the model**.

Comparison with Prompt Engineering

Prompt engineering improved the behavior of the same LLM by giving it detailed instructions about:

- accuracy,
- response structure,
- conciseness,
- formatting,
- uncertainty, and
- avoidance of unsupported claims.

Lowering the temperature from 0.7 to 0.2 also made the prompt-engineered responses more focused and deterministic.

However, prompt engineering still did not provide the model with the assigned HBR document.

This created an important limitation:

Prompt engineering improved how the model answered, but RAG improved what evidence the model could use to answer.

RAG retains the advantages of a carefully designed prompt while also adding retrieved source material.

Comparison of the Three Approaches

Capability	Baseline LLM	Prompt-Engineered LLM	RAG
Uses pretrained model knowledge	Yes	Yes	Yes, but constrained by retrieved context
Uses the assigned HBR document	No	No	Yes
Structured instructions	Minimal	Strong	Strong

Semantic retrieval	No	No	Yes
Vector database	No	No	Yes
Source-page traceability	No	No	Yes
Document-specific grounding	Low	Low	High
Hallucination risk	Higher	Reduced somewhat	Lower when relevant context is retrieved
Ability to verify claims	Limited	Limited	Stronger
Suitable for enterprise document analysis	Limited	Moderate	Strong

How RAG Improves the Three Project Questions

Question 1 — Authors and publisher

The baseline and prompt-engineered models could potentially answer this from prior knowledge, but RAG can retrieve the article's title and publication information directly from the source PDF. This makes the answer verifiable rather than merely plausible.

Question 2 — Three leadership characteristics

This question shows particularly strong retrieval performance. The vector search retrieved the passage that explicitly lists the three characteristics requested by the question. This demonstrates that semantic retrieval can identify a precise answer embedded within a larger business report.

Question 3 — Leadership and successful innovation

This is a more complex analytical question because the answer may require information from multiple passages. RAG can retrieve examples such as Apple's functional structure, expert-led decision rights, detailed attention to products, and the dual-lens camera/portrait-mode decision. The LLM can then synthesize those passages into a concise explanation while remaining grounded in the report.

Overall RAG Assessment

The RAG implementation provides the strongest approach of the three tested methods because it combines:

- semantic retrieval,

- prompt engineering,
- source-document context,
- deterministic generation,
- metadata preservation, and
- natural-language response generation.

The resulting architecture is better suited to business-document analysis because the model is no longer expected to recall every relevant fact from training. Instead, the system retrieves relevant evidence when it is needed and uses the LLM primarily to interpret, summarize, and communicate that evidence.

This directly supports the project objective of reducing the amount of time analysts spend manually searching long reports while improving the reliability and usefulness of the resulting insights.

Important Limitations

RAG improves factual grounding, but it does **not eliminate all risk of error**.

The quality of a RAG response depends heavily on the quality of retrieval. If the correct passage is not retrieved, the model cannot reliably use information that is absent from its context.

The similarity-search output also showed some duplicated or highly overlapping chunks. Although overlap helps preserve context, duplicate retrieval can consume context-window space without adding new information. A production implementation could improve this by:

- deduplicating retrieved chunks,
- testing different values of `k`,
- using Maximum Marginal Relevance (MMR),
- adding metadata filters, or
- reranking retrieved passages before sending them to the LLM.

Therefore, RAG should be viewed as a major improvement in evidence-based generation, but not as an automatic guarantee of correctness.

Actionable Insights and Business Recommendations

Actionable Insights and Business

Recommendations

Actionable Insights

1. RAG can significantly reduce manual document-review effort

The prototype demonstrates that semantic retrieval can locate relevant passages within a lengthy business report without requiring the analyst to manually read the document from beginning to end.

For organizations reviewing large volumes of research reports, market studies, investment documents, or internal reports, this could substantially reduce information-search time and allow analysts to focus more effort on interpretation and decision-making.

2. Source traceability is critical for business use

The most important advantage of the RAG approach is not simply faster answers; it is the ability to connect generated responses back to the original source.

Preserving PDF page metadata allows analysts to verify important claims before using them in an investment recommendation or strategic decision.

For high-impact business workflows, traceability should be treated as a core design requirement rather than an optional feature.

3. Prompt engineering and RAG solve different problems

The project demonstrates that prompt engineering and RAG are complementary.

Prompt engineering improves:

- response structure,
- instruction-following,
- consistency,
- conciseness, and
- communication quality.

RAG improves:

- source grounding,
- document-specific factual accuracy,
- retrieval of relevant evidence,
- traceability, and
- verifiability.

A production application should therefore combine both approaches rather than treating them as competing alternatives.

4. Retrieval quality determines downstream answer quality

The language model can only reason effectively over the information that has been successfully retrieved.

Therefore, document chunking, embeddings, retrieval strategy, the value of `k`, metadata quality, and passage reranking should be treated as important model-design parameters.

The retrieval stage should be evaluated independently rather than assuming that a fluent final answer means that retrieval was successful.

5. Apple's organizational model provides a strategic leadership insight

The retrieved article suggests that Apple connects **expertise with decision rights** rather than relying primarily on traditional general-management structures.

The report identifies three important leadership behaviors:

- deep expertise,
- immersion in details, and
- collaborative debate.

For organizations operating in highly technical or rapidly changing markets, this suggests that decision authority may be more effective when placed closer to domain expertise.

6. Innovation may require protecting long-term product decisions from excessive short-term financial pressure

The retrieved content describes how Apple's functional organization enables technical leaders to weigh user value and innovation against cost rather than allowing short-term product-level financial targets to dominate every decision.

The iPhone 7 Plus dual-lens camera and portrait-mode example illustrates how expert-led organizations may be more willing to make technically informed investments whose value is uncertain at the time of development.

Business Recommendations

Recommendation 1 — Deploy RAG for research-intensive analyst workflows

Organizations that regularly analyze lengthy documents should consider RAG-based assistants for:

- investment research,
- competitive intelligence,
- market reports,
- earnings documents,
- technical reports,
- due-diligence materials, and
- internal knowledge repositories.

The goal should be to augment analyst judgment rather than replace it.

Recommendation 2 — Preserve metadata and citations throughout the pipeline

Every document ingested into an enterprise RAG system should retain metadata such as:

- document name,
- page number,
- section,
- publication date, and
- source location.

Generated responses should surface this metadata so analysts can verify important claims.

Recommendation 3 — Use human validation for high-impact decisions

RAG improves grounding but should not be treated as an autonomous decision-maker.

For investment, legal, financial, or strategic recommendations, users should verify material conclusions against the source documents before acting.

The system should therefore be designed as a **decision-support tool**, not as a replacement for expert judgment.

Recommendation 4 — Measure retrieval performance separately from generation quality

Organizations should evaluate whether the retriever is returning the correct passages before evaluating the quality of the final LLM response.

Useful evaluation measures could include:

- retrieval precision,
- retrieval recall,
- answer relevance,
- faithfulness to context,
- citation accuracy, and
- completeness.

A fluent response should not automatically be interpreted as an accurate response.

Recommendation 5 — Optimize retrieval before scaling

The current prototype uses similarity search with `k=4`, which performed well for the leadership-characteristics question.

Before deploying the system at enterprise scale, the organization should experiment with:

- different chunk sizes,
- different overlap values,
- alternative embedding models,
- different values of `k`,
- MMR retrieval,
- reranking, and
- duplicate-chunk removal.

These changes could improve retrieval diversity and reduce irrelevant or repetitive context.

Recommendation 6 — Extend the prototype to multiple documents

The current solution analyzes a single HBR article. A production implementation should support collections of documents so analysts can ask questions across multiple reports simultaneously.

Additional capabilities could include:

- document filtering,
- date filtering,
- company filtering,
- source ranking,
- automatic citations,

- conversation history, and
- side-by-side source comparison.

Overall Business Recommendation

The prototype demonstrates that RAG is a practical approach for reducing information overload in document-intensive business environments.

The greatest business value comes from combining the natural-language capabilities of an LLM with **retrieval, provenance, and human verification**.

For a venture-capital firm or similar research-driven organization, this approach could shorten the time required to review large collections of reports while allowing analysts to spend more time evaluating strategic implications, identifying patterns, and making higher-value decisions.

Power Ahead
